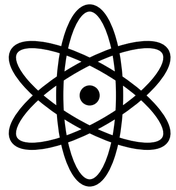
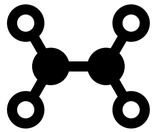
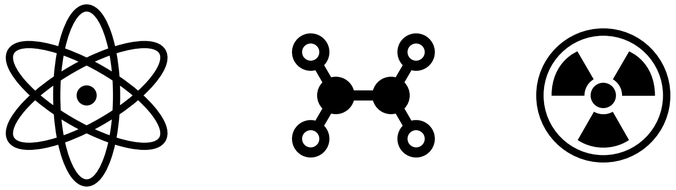
Deep Learning for forecasting quantum dynamics


Quantum mechanics describes the behavior of matter at microscopic level: ***atoms, molecules, materials, radiation, fundamental particles*** etc.

Deep Learning for forecasting quantum dynamics

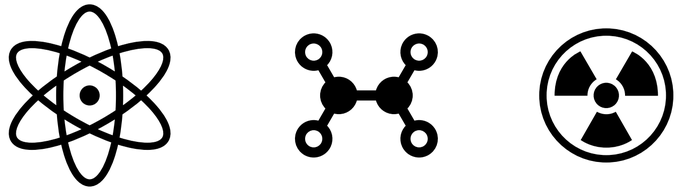


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The understanding of microscopic (or quantum) matter has led to some of the most important technological innovations of the past century, such as ***most modern electronic devices (e.g., transistors), lasers, magnetic resonance imaging, atomic clocks,*** and so on.

New quantum technologies currently under development include: quantum sensors, quantum communication, and ***quantum simulation & computation***.

Deep Learning for forecasting quantum dynamics



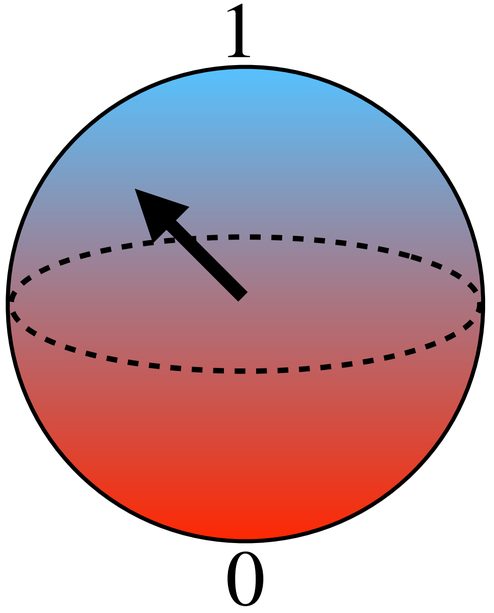
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The fundamental unit of information in quantum mechanics is the ***qubit***, which, unlike the classical bit, can take on both values, ***0 and 1 simultaneously*****,** thanks to the ***principle of superposition***.

The ***principle of superposition*** is the fundamental concept underlying ***quantum technologies***.



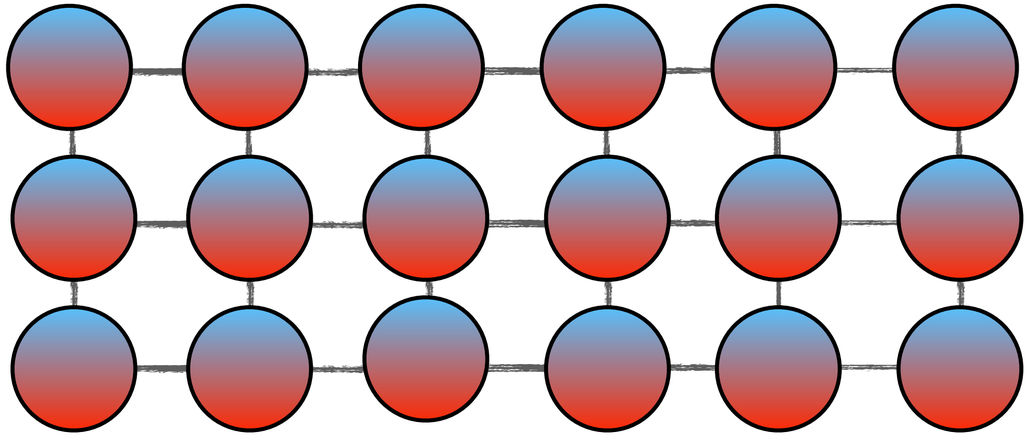
Graphical representation of a **qubit**

Deep Learning for forecasting quantum dynamics

Qubits mathematically describe the elements of microscopic matter, such as **atoms**.

Systems composed of many interacting **qubits** are characterized by peculiar phenomena, such as **entanglement**.

Their understanding is the key to the development of **new technologies** that exploit **purely quantum effects**.

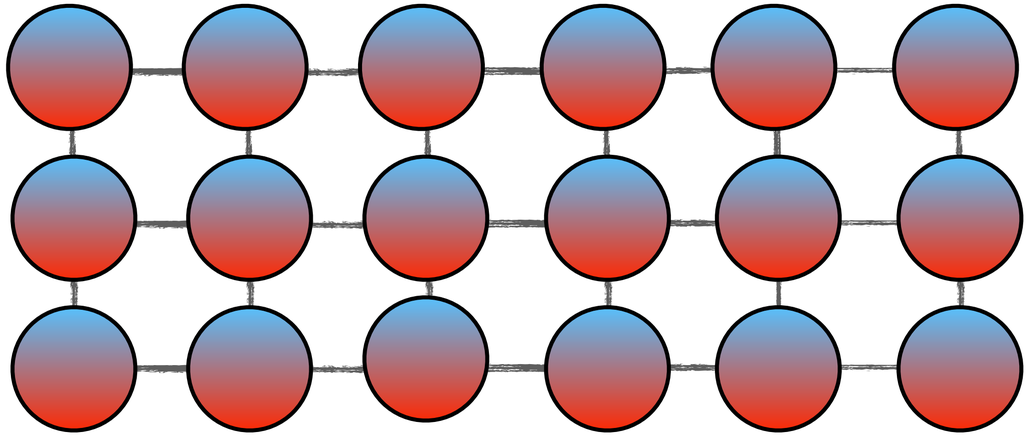


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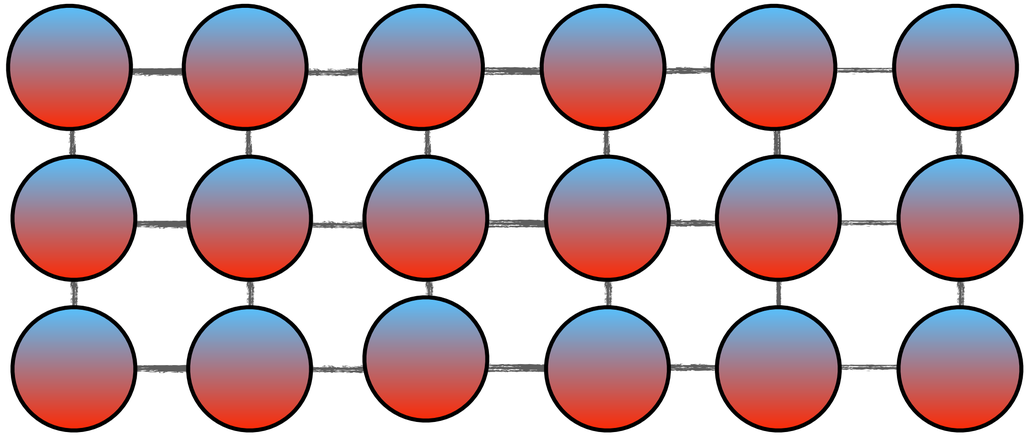


Traditional methods used to study them can solve systems with **small number of qubits** or can simulate their behavior only for **short times**

Overcoming these limitations is of crucial importance.

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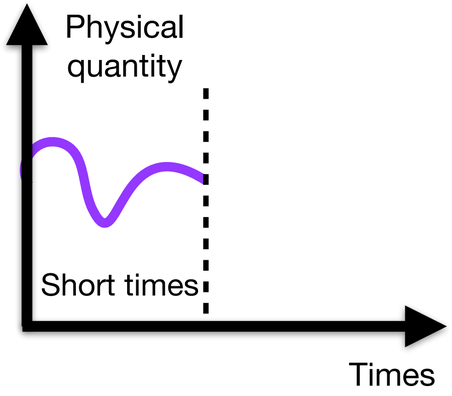
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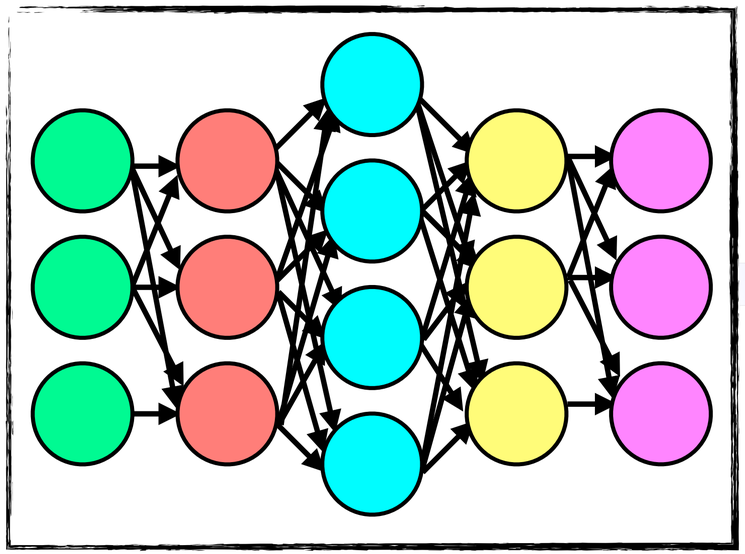
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Deep learning represents a promising solution to these problems

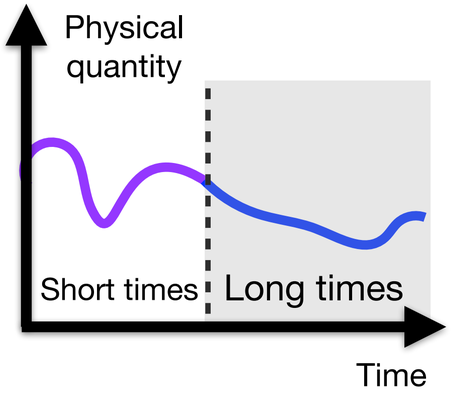
Quantum dynamics from traditional simulations (Input data)



NEURAL NETWORK



Forecasting the dynamics with deep learning (output data)



Deep Learning for forecasting quantum dynamics

Technological relevance:

Several companies, including **Google, IBM, Amazon, Microsoft** (and many other specialized companies and startups), have invested in quantum technologies.

The **quantum properties of matter** are expected to provide extreme advantages in computation for solving specific tasks, such as in **cryptography.**

Understanding the **dynamics of quantum systems**, such as atoms and molecules, is fundamental for the design of **new materials**. **Quantum mechanics** also predicts the existence of **new phases of matter** (e.g., superconductivity), which offer promising applications in **sensing, quantum computer components, particle accelerators, communication**, and more.

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Scientific relevance:

The last four **Nobel Prizes in Physics** have been awarded for work on the study of **quantum systems** (2022, 2023, 2025) and **machine learning** (2024).

The study of the **dynamics of complex quantum systems** helps to shed light on the **fundamental properties of the microscopic world**. Over the years, it has led to the discovery of extraordinary phenomena such as **entanglement, tunneling**, and more. Understanding the microscopic world corresponds to understanding the behavior of the **smallest building blocks of matter**, such as fundamental particles including **protons, neutrons, quarks**, and others.